

Bitcoin Market Sentiment vs Hyperliquid Trader Performance

1. Introduction

Financial markets are strongly influenced by investor psychology. The Bitcoin Fear & Greed Index is a popular indicator used to measure market sentiment by categorizing market conditions into Fear, Neutral, Greed, and Extreme Greed.

This project combines sentiment data with Hyperliquid trading records to analyze how trader performance changes under different market sentiment conditions.

Objectives

- Analyze trader profitability across sentiment regimes.
- Investigate the relationship between sentiment and trading performance.
- Examine directional trading behavior (BUY vs SELL).
- Evaluate the impact of position size on profitability.
- Generate actionable trading insights.

2. Dataset Overview

Dataset 1: Bitcoin Fear & Greed Index

The sentiment dataset contains:

- Date
- Sentiment Classification
- Sentiment Value

The sentiment categories include:

- Fear
- Neutral
- Greed
- Extreme Greed

Dataset 2: Hyperliquid Historical Trading Data

The trading dataset contains:

- Account
- Coin

- Execution Price
- Size USD
- Side (BUY / SELL)
- Closed PnL
- Timestamp

The trading data was merged with the sentiment dataset using transaction dates.

3. Methodology

The analysis was performed in Python using Pandas, Matplotlib, and Seaborn.

Data Preparation

The following preprocessing steps were performed:

1. Converted timestamps into date format.
2. Cleaned and standardized profit and loss values.
3. Merged sentiment and trading datasets using trading dates.
4. Created additional metrics such as:
 - Win Rate
 - Average Profit per Trade
 - Total Profitability

Analytical Techniques

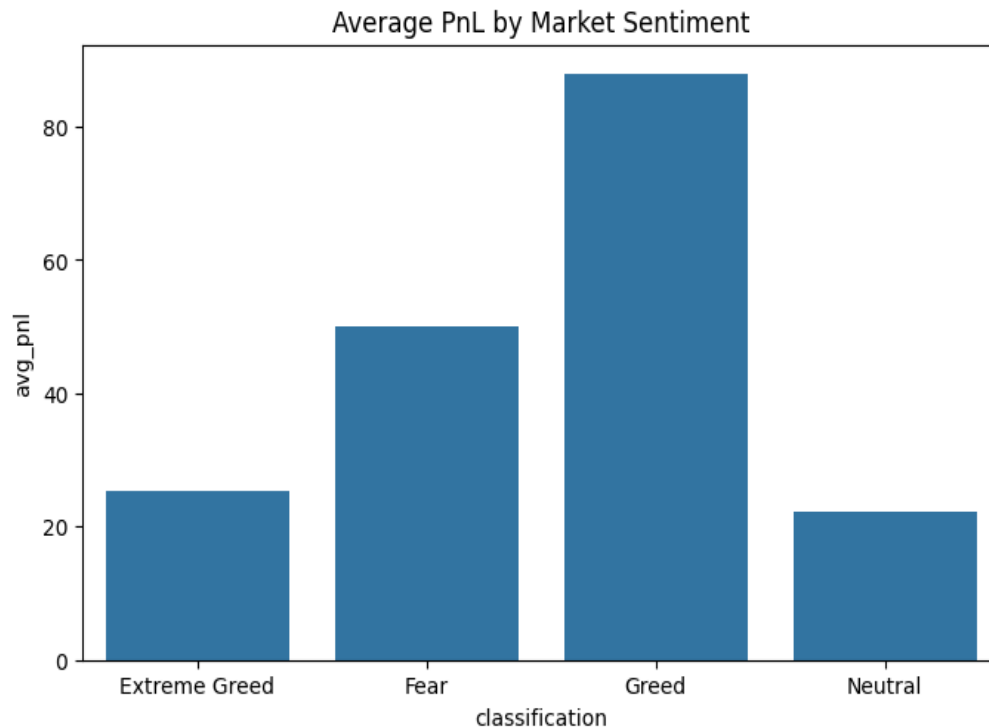
The following analyses were conducted:

- Sentiment Analysis
- Trader Performance Analysis
- Trade Direction Analysis
- Position Size Analysis
- Correlation Analysis
- Daily Profitability Analysis

4. Analysis and Findings

4.1 Profitability Across Market Sentiment

Average profitability varied significantly across sentiment regimes.



Results

Sentiment	Average PnL
Greed	87.89
Fear	50.05
Extreme Greed	25.42
Neutral	22.23

Observation

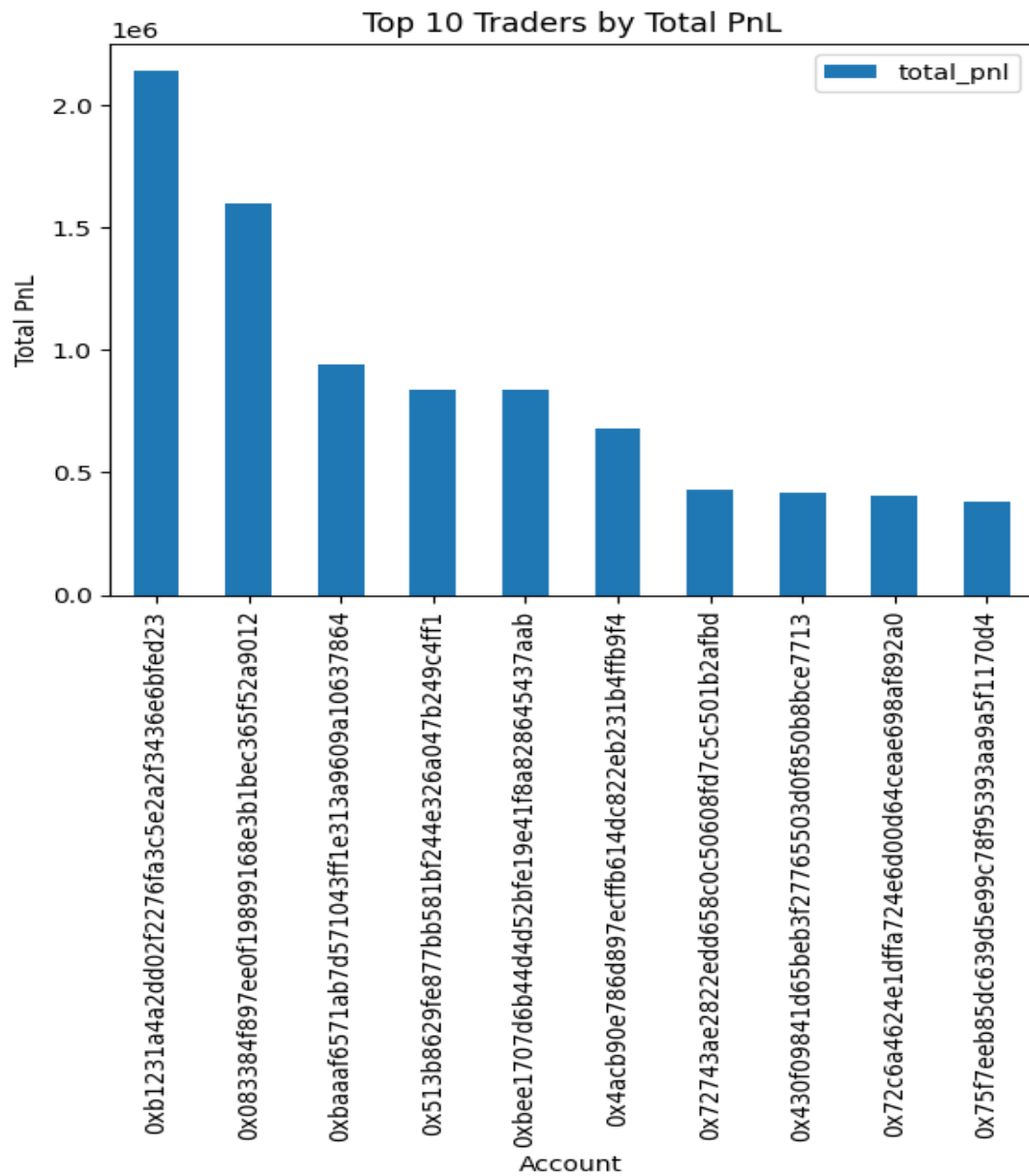
Greed periods generated the highest average profit per trade. Fear periods also produced strong profitability but were associated with substantially higher trading activity.

Interpretation

Bullish market conditions appear to create favorable opportunities for traders. However, fear-driven markets also provide significant opportunities due to increased volatility.

4.2 Top Trader Analysis

A small number of traders accounted for a large proportion of overall profitability.



Observation

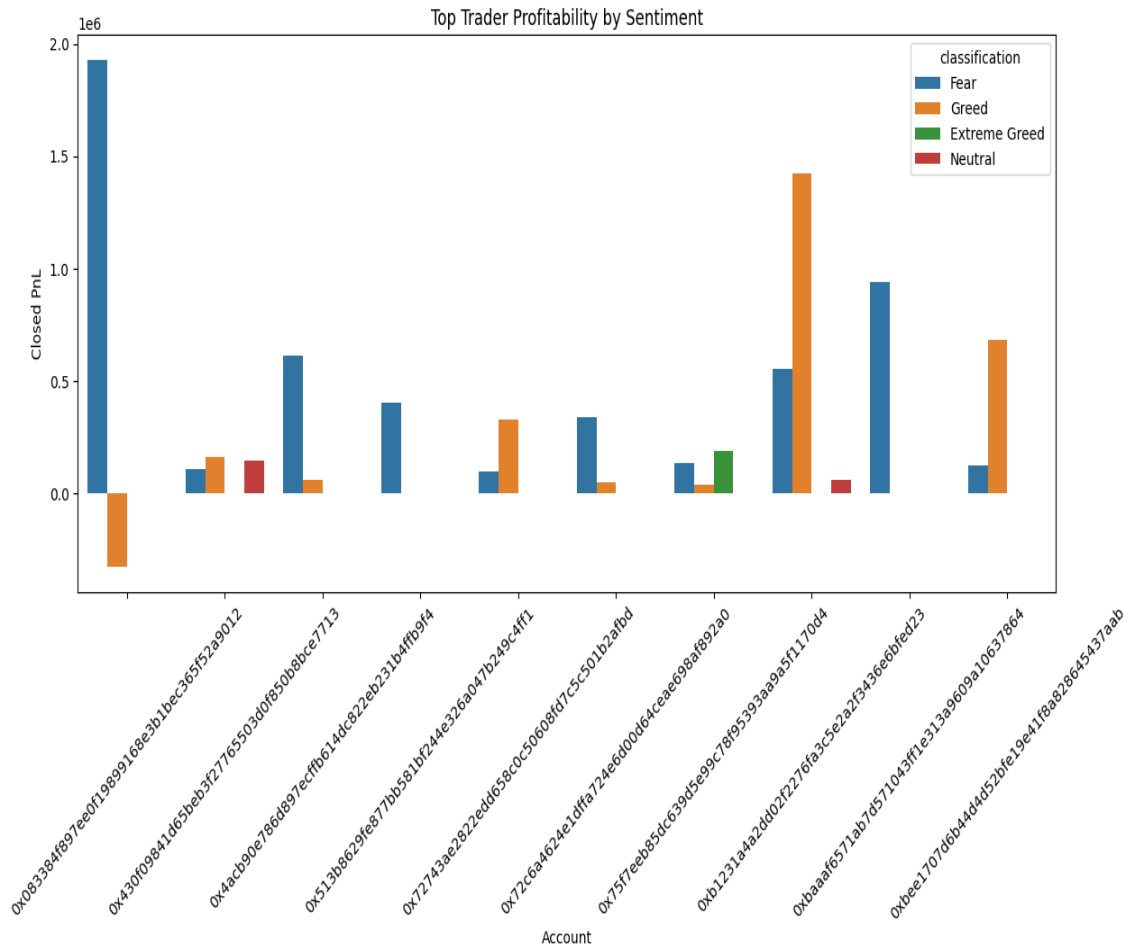
The top-performing accounts generated significantly higher profits than the rest of the trader population.

Interpretation

Profitability is concentrated among a small subset of experienced traders, indicating substantial differences in trading skill and strategy effectiveness.

4.3 Trader Performance by Sentiment

Analysis of the most profitable traders revealed varying performance patterns across sentiment regimes.



Observation

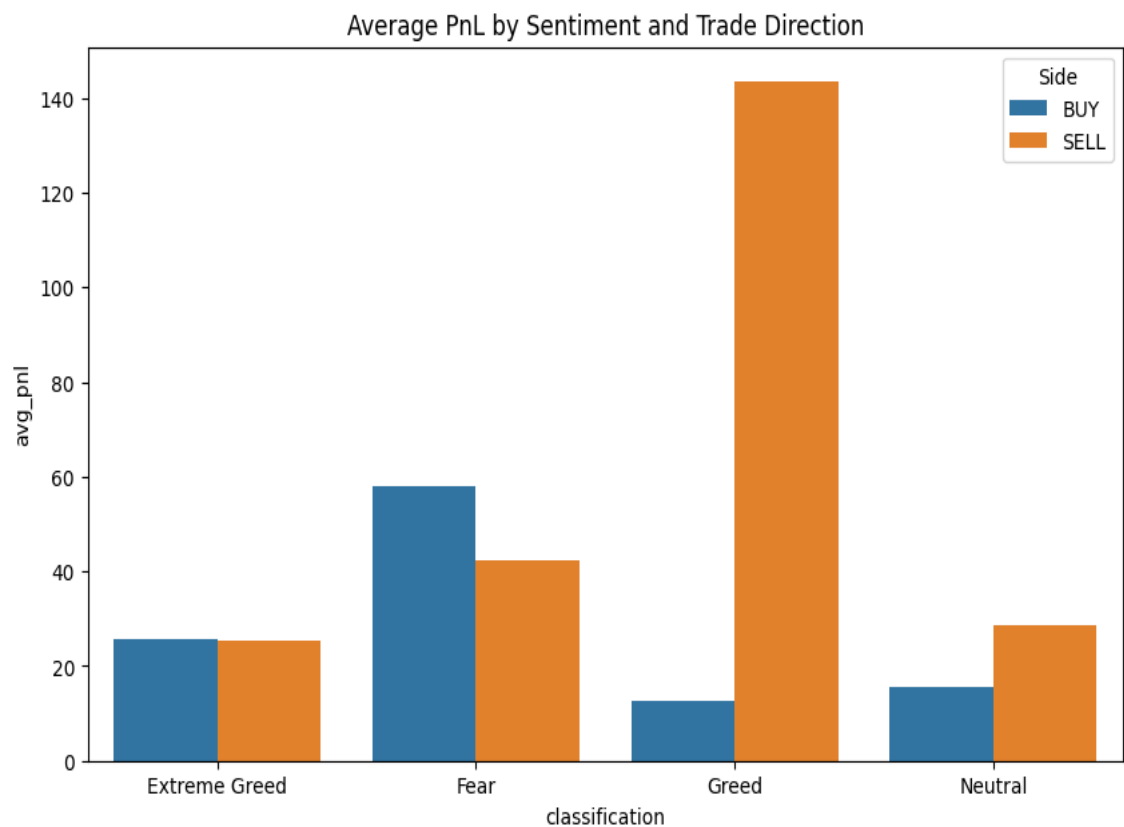
Some traders generated most of their profits during Fear periods, while others performed better during Greed periods.

Interpretation

Successful traders adapt their strategies according to prevailing market conditions rather than relying on a single approach.

4.4 Trade Direction Analysis

Average profitability was analyzed separately for BUY and SELL trades.



Results

Sentiment	BUY Avg PnL	SELL Avg PnL
Fear	58.07	42.22
Greed	12.49	143.62
Neutral	15.65	28.58
Extreme Greed	25.58	25.27

Observation

BUY trades outperformed SELL trades during Fear markets, while SELL trades significantly outperformed BUY trades during Greed markets.

Interpretation

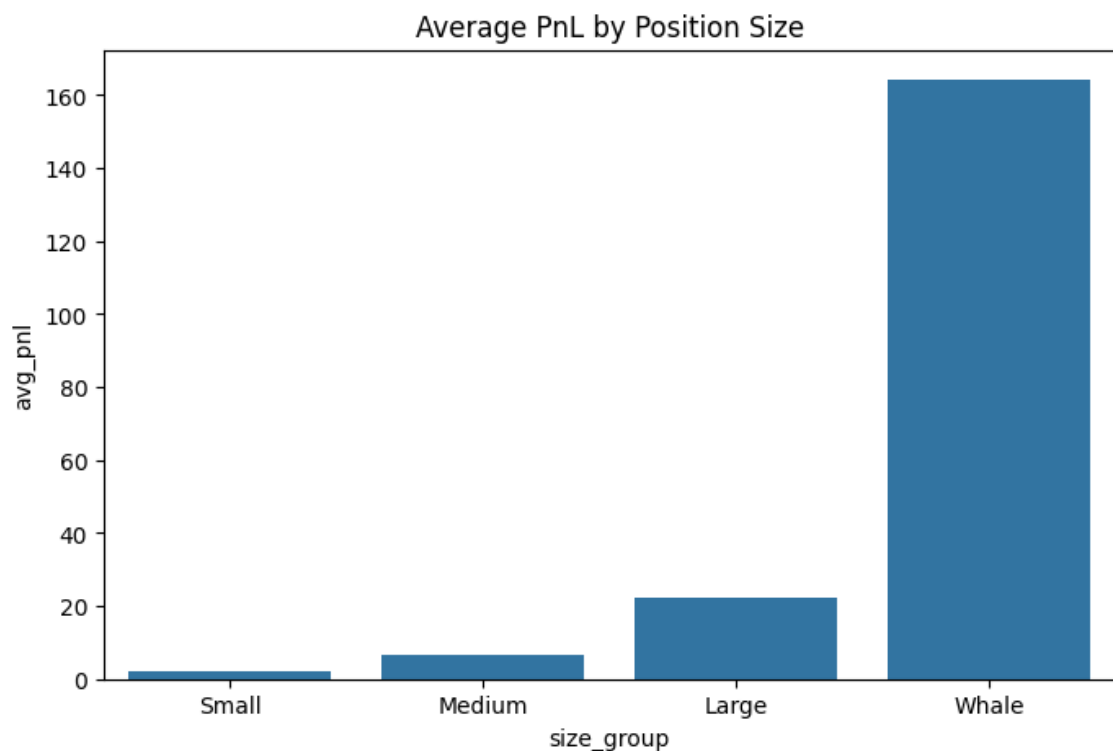
This behavior suggests that profitable traders frequently adopt contrarian strategies:

- Buying during fearful conditions
- Selling during optimistic conditions

4.5 Position Size Analysis

Traders were segmented into four groups based on position size:

- Small
- Medium
- Large
- Whale



Observation

Average profitability increased consistently with position size.

Interpretation

Larger positions were associated with significantly higher profits, suggesting that capital allocation and risk management play an important role in trading success.

4.6 Sentiment Correlation Analysis

A numerical sentiment score was created and compared with individual trade profitability.

Result

Correlation between sentiment score and Closed PnL:

0.009

Interpretation

The relationship is extremely weak.

Although profitability differs across sentiment regimes, sentiment alone is not sufficient to predict the outcome of individual trades.

4.7 Daily Profitability Analysis

Daily trading profits were compared against the Fear & Greed score.



Observation

Profitability varied across sentiment levels, but no strong linear relationship was observed.

Interpretation

Market sentiment influences the broader trading environment but should not be treated as a standalone predictive indicator.

5. Key Insights

Insight 1

Greed periods generated the highest average profit per trade.

Insight 2

Fear periods produced the highest trading activity and total market participation.

Insight 3

Trader performance is highly dependent on market sentiment regimes.

Insight 4

Contrarian trading strategies appear highly effective.

- BUY during Fear
- SELL during Greed

Insight 5

Larger position sizes generated substantially higher profitability.

Insight 6

Market sentiment alone exhibited weak predictive power for individual trade outcomes.

6. Trading Recommendations

Recommendation 1

Use market sentiment as a market-regime classifier rather than a direct trading signal.

Recommendation 2

Consider contrarian positioning during extreme sentiment conditions.

Recommendation 3

Implement dynamic position sizing based on prevailing market conditions.

Recommendation 4

Combine sentiment indicators with trader behavior, position size, and trade direction for stronger decision-making.

Recommendation 5

Monitor trader-specific performance across sentiment regimes to identify strategy specialization.

7. Conclusion

This study explored the relationship between Bitcoin market sentiment and trader performance using Hyperliquid trading data and the Fear & Greed Index.

The results demonstrate that sentiment influences aggregate profitability and trading behavior, but sentiment alone is not a reliable predictor of individual trade outcomes. The strongest performance patterns emerged when sentiment was combined with trade direction and position sizing.

Overall, sentiment should be viewed as contextual market information that can enhance trading decisions when integrated with broader market and behavioral indicators.

Future work could include leverage analysis, risk-adjusted return metrics, and machine learning models for trader classification and profitability prediction.